A QUOTIENT OF A SOBER SPACE BY AN EQUIVALENCE RELATION WITH FINITE FIBRES AND LIFTING GENERALIZATIONS IS KOLMOGOROV

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Allowed standard prerequisites (editorial).

- The Kolmogorov separation axiom (distinct points are separated by an open set one way or the other) and the specialization relation u leads to u-prime between points. Set up in the author's sections on specialization and on Kolmogorov spaces, before the counted statement. Fixes the conclusion of the statement and the relation used throughout the proof.
- The hypotheses of the statement: openness of the map, sobriety of the source, finiteness of the fibres of the two maps, lifting of generalizations along them, and the quotient description of the target. The five numbered items of the counted statement itself (source lines 3936-3943). Each hypothesis is used in the proof: finiteness makes equivalence classes finite, lifting of generalizations transfers specializations, and openness relates the topology of the quotient to that of the source.

Lemma 0.1. *Suppose that $s, t : R \to U$ and $\pi : U \to X$ are continuous maps of topological spaces such that*

(1) *π is open,*
(2) *U is sober,*
(3) *s, t have finite fibres,*
(4) *generalizations lift along s, t,*

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(5) $(t, s)(R) \subset U \times U$ *is an equivalence relation on* U *and* X *is the quotient of* U *by this equivalence relation (as a set).*

Then X *is Kolmogorov.*

Proof. Properties (3) and (5) imply that a point x corresponds to an finite equivalence class $\{u_1, \ldots, u_n\} \subset U$ of the equivalence relation. Suppose that $x' \in X$ is a second point corresponding to the equivalence class $\{u'_1, \ldots, u'_m\} \subset U$. Suppose that $u_i \rightsquigarrow u'_j$ for some i, j. Then for any $r' \in R$ with $s(r') = u'_j$ by (4) we can find $r \rightsquigarrow r'$ with $s(r) = u_i$. Hence $t(r) \rightsquigarrow t(r')$. Since $\{u'_1, \ldots, u'_m\} = t(s^{-1}(\{u'_j\}))$ we conclude that every element of $\{u'_1, \ldots, u'_m\}$ is the specialization of an element of $\{u_1, \ldots, u_n\}$. Thus $\overline{\{u_1\}} \cup \ldots \cup \overline{\{u_n\}}$ is a union of equivalence classes, hence of the form $\pi^{-1}(Z)$ for some subset $Z \subset X$. By (1) we see that Z is closed in X and in fact $Z = \overline{\{x\}}$ because $\pi(\overline{\{u_i\}}) \subset \overline{\{x\}}$ for each i. In other words, $x \rightsquigarrow x'$ if and only if some lift of x in U specializes to some lift of x' in U, if and only if every lift of x' in U is a specialization of some lift of x in U.

Suppose that both $x \rightsquigarrow x'$ and $x' \rightsquigarrow x$. Say x corresponds to $\{u_1, \ldots, u_n\}$ and x' corresponds to $\{u'_1, \ldots, u'_m\}$ as above. Then, by the results of the preceding paragraph, we can find a sequence

$$\ldots \rightsquigarrow u'_{j_3} \rightsquigarrow u_{i_3} \rightsquigarrow u'_{j_2} \rightsquigarrow u_{i_2} \rightsquigarrow u'_{j_1} \rightsquigarrow u_{i_1}$$

which must repeat, hence by (2) we conclude that $\{u_1, \ldots, u_n\} = \{u'_1, \ldots, u'_m\}$, i.e., $x = x'$. Thus X is Kolmogorov. □